Chapter 5: Thermochemistry

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry

Southern Union State Community College

PHY201

Lecture 5

Part I

Energy Basics

Outline of Part I: Energy Basics

Outline of Part I: Energy Basics

What Is Energy?

The First Law of Thermodynamics

Work — Pressure-Volume (PV) Work

Energy

Definition: Energy

Energy is the capacity to do work or to produce heat.

Kinetic Energy

Energy of **motion**.

Examples:

- A moving car
- Flowing water
- Vibrating atoms in a warm object

Potential Energy

Energy of **position** or stored energy.

Examples:

- Water held behind a dam
- A compressed spring
- Chemical bonds in fuel

In chemistry, the most important form of potential energy is **chemical energy** — energy stored in chemical bonds.

Forms of Energy

Energy exists in many forms, all of which can be converted into one another:

- **Mechanical:**
  energy of moving objects or stored in stretched/compressed materials
- **Radiant (electromagnetic):**
  energy carried by light and other radiation
- **Nuclear:**
  energy stored in atomic nuclei
- **Electrical:**
  energy from moving electric charges
- **Chemical:**
  energy stored in chemical bonds; released or absorbed during reactions
- **Thermal:**
  total kinetic energy of all particles in a sample

📝 The **transfer of energy** from one form to another appears as **work** and/or **heat**.

Energy Units

The SI unit of energy is the **joule** (J):

$$1\,\mathrm{J} = 1\,\mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}^2$$

The Calorie

- **calorie (cal)**: amount of heat needed to raise 1 g of water by 1 °C.
- **Calorie (Cal)** = 1 kcal = food energy unit

$$1\,\mathrm{cal} = 4.184\,\mathrm{J}$$

$$1\,\mathrm{Cal} = 1\,\mathrm{kcal} = 4184\,\mathrm{J}$$

A cheeseburger with 500 Cal contains

$$500\,\mathrm{Cal} \times \frac{4184\,\mathrm{J}}{1\,\mathrm{Cal}} \approx 2.1 \times 10^6\,\mathrm{J}$$

of chemical potential energy.

Check Your Understanding

Classify each as primarily **kinetic** or **potential** energy:

(a) A ball rolling down a ramp

(b) Gasoline sitting in a fuel tank

(c) A stretched rubber band

(d) Steam flowing through a pipe

Come to lecture to see the answer.

Outline of Part I: Energy Basics

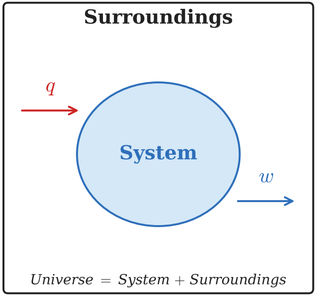
System and Surroundings

System

The part of the universe under study — usually the reacting chemicals.

Surroundings

Everything outside the system — the container, the room, etc.



Energy can be exchanged between system and surroundings in two ways:

- **Heat** (q): energy transferred due to a temperature difference
- **Work** (w): energy transferred by a force acting through a distance

The **universe** is defined as any system plus its surroundings.

Internal Energy and the First Law

Internal Energy (E or U)

The **internal energy** of a system is the sum of all its particles' kinetic and potential energies.

First Law of Thermodynamics (1LOT)

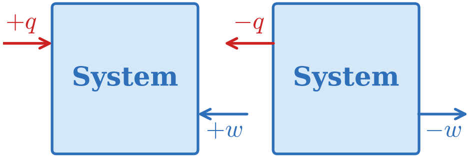
Energy is **neither created nor destroyed**; it is simply transferred from one form to another.

$$\Delta E_{\text{universe}} = \Delta E_{\text{system}} + \Delta E_{\text{surroundings}} = 0$$

📝 The change in internal energy of a system equals the heat added to it plus the work done on it:

$$\Delta E = q + w$$

Sign Conventions for q and w



$\Delta E > 0$
(System's E $\uparrow$)

$\Delta E < 0$
(System's E $\downarrow$)

$$\Delta E = q + w$$

Heat (q)

- $q > 0$: heat flows **into** the system (endothermic)
- $q < 0$: heat flows **out of** the system (exothermic)

Work (w)

- $w > 0$: work done **on** the system (surroundings compress system)
- $w < 0$: work done **by** the system (system expands against surroundings)

Check Your Understanding

A chemical reaction releases heat to the surroundings and causes a gas to expand against a piston.

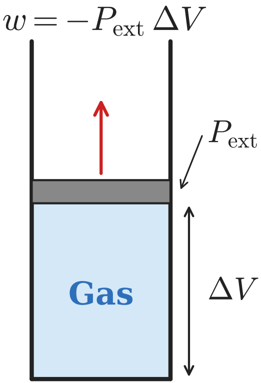
What are the signs of q and w for the **system**?

Is ΔE positive or negative? Is it large or small relative to q alone?

Come to lecture to see the answer.

Outline of Part I: Energy Basics

Pressure–Volume (PV) Work



Expansion: $\Delta V > 0 \Rightarrow w < 0$

PV Work

When a gas expands or contracts against a constant external pressure P, the work done *by* the system (the gas) is:

$$w = -P\,\Delta V$$

where $\Delta V = V_{\text{final}} - V_{\text{initial}}$.

Expansion ($\Delta V > 0$)

Gas pushes piston out.

$$w = -P\Delta V < 0$$

System *loses* energy by doing work on surroundings.

Contraction ($\Delta V < 0$)

Piston compresses gas.

$$w = -P\Delta V > 0$$

System *gains* energy as surroundings do work on it.

Changing the Amount of Gaseous Species Changes Volume

Expansion ($\Delta V > 0$): More moles of gas produced

$$\mathrm{C_3H_8(g)} + 5\,\mathrm{O_2(g)} \longrightarrow 3\,\mathrm{CO_2(g)} + 4\,\mathrm{H_2O(g)}$$

$$6 \text{ mol gas} \to 7 \text{ mol gas} \therefore \Delta V > 0,\ \ w < 0$$

Contraction ($\Delta V < 0$): Fewer moles of gas produced

$$3\,\mathrm{H_2(g)} + \mathrm{N_2(g)} \longrightarrow 2\,\mathrm{NH_3(g)}$$

$$4 \text{ mol gas} \to 2 \text{ mol gas} \therefore \Delta V < 0,\ \ w > 0$$

When counting moles of gas, **only** gas-phase species matter — solids and liquids do not contribute significantly to ΔV.

Check Your Understanding

For each reaction, predict the sign of w (work done on/by the system):

(a) $2\,SO_2(g) + O_2(g) \longrightarrow 2\,SO_3(g)$

(b) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

(c) $N_2(g) + 3\,H_2(g) \longrightarrow 2\,NH_3(g)$

Come to lecture to see the answer.

Part II

Calorimetry

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Heat Capacity

Heat Capacity (C)

The **heat capacity** of an object is the quantity of heat required to raise its temperature by 1 °C (or 1 K).

$$q = C\Delta T$$

Heat capacity is an **extensive** property — it depends on the amount of material.

Specific Heat Capacity (c or c_s)

The **specific heat capacity** is the heat required to raise 1 g of a substance by 1 °C. It is an **intensive** property.

$$q = m\,c\,\Delta T$$

Units: $\frac{\mathrm{J}}{\mathrm{g\,^\circ C}}$ or $\frac{\mathrm{J}}{\mathrm{g\,K}}$

Specific Heat Capacities of Common Substances

Substance	**Specific Heat (c), $\frac{J}{g\,°C}$**
Water (liquid)	4.184
Water (ice)	2.093
Water (steam)	1.864
Aluminum	0.897
Iron	0.449
Copper	0.385
Gold	0.129

Water has an exceptionally **high** specific heat — it resists temperature change. This is why coastal climates are mild and why water is used in cooling systems.

Check Your Understanding

Two blocks of equal mass — one aluminum ($c = 0.897\,\frac{\text{J}}{\text{g}\,^\circ\text{C}}$) and one gold ($c = 0.129\,\frac{\text{J}}{\text{g}\,^\circ\text{C}}$) — are heated with the same amount of energy.

Which block reaches a higher temperature? Why?

Come to lecture to see the answer.

Example: Using $q = mc\,\Delta T$

Heating Iron

How much heat is necessary to raise the temperature of 1.00 g of iron from 17.0 °C to 27.0 °C?
$\left(c_{Fe} = 0.449\,\frac{J}{g\,°C}\right)$

Given:

$$m = 1.00\,g$$
$$c = 0.449\,\frac{J}{g\,°C}$$
$$\Delta T = 27.0 - 17.0 = 10.0\,°C$$

$$q = mc\,\Delta T$$
$$q = (1.00\,g)(0.449\,\frac{J}{g\,°C})(10.0\,°C)$$
$$\boxed{q = 4.49\,J}$$

Check Your Understanding

A $55.0\,\mathrm{g}$ piece of copper $\left(c = 0.385\,\frac{\mathrm{J}}{\mathrm{g}\,^\circ\mathrm{C}}\right)$ is heated from $25.0\,^\circ\mathrm{C}$ to $75.0\,^\circ\mathrm{C}$.

How much heat does the copper absorb?

Come to lecture to see the answer.

Outline of Part II: Calorimetry

Heat Capacity and Specific Heat

Calorimetry

Calorimetry

Definition: Calorimetry

Calorimetry is the measurement of heat flow associated with a chemical or physical process.

Calorimeter

A **calorimeter** is an insulated device used to measure heat changes. Because the calorimeter is insulated, all heat from the reaction stays within the system:

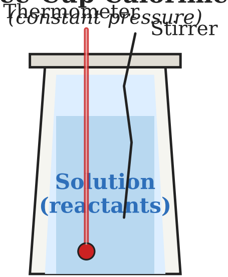
$$q_{\text{rxn}} = -q_{\text{calorimeter}}$$

The negative sign is because the heat *released by* the reaction equals the heat *absorbed by* the calorimeter (and vice versa).

Coffee-Cup Calorimetry

- Simple, open to atmosphere — measures at **constant pressure**.
- $q_{\text{rxn}} = -q_{\text{solution}} = -mc_s\Delta T$
- We assume $c_{\text{solution}} \approx c_{\text{water}} = 4.184\,\frac{\text{J}}{\text{g}\,^\circ\text{C}}$
- Measures **enthalpy change** (ΔH) directly.

Coffee-Cup Calorimeter
(constant pressure)



$$q_{\text{rxn}} = -q_{\text{soln}} = -mc_s\Delta T$$

Sign check: if T goes up, reaction is exothermic ($q_{\text{rxn}} < 0$).

Example: Coffee-Cup Calorimetry

Dissolving NaOH

5.00 g of NaOH is dissolved in 100.0 g of water at 23.6 °C. The final temperature is 47.4 °C. Find q_{rxn}.

Given:

$$m = 100.0 + 5.00 = 105.0\,\text{g}$$
$$c = 4.184\,\frac{\text{J}}{\text{g}\,^\circ\text{C}}$$
$$\Delta T = 47.4 - 23.6 = 23.8\,^\circ\text{C}$$

$$\begin{aligned} q_{\text{soln}} &= mc\,\Delta T \\ &= (105.0)(4.184)(23.8) \\ &= 10\,450\,\text{J} \end{aligned}$$

$$\boxed{q_{\text{rxn}} = -10\,450\,\text{J}}$$

$$\boxed{q_{\text{rxn}} \approx -10.5\,\text{kJ}}$$

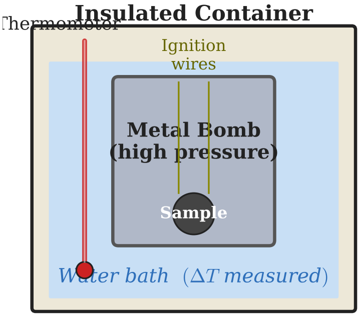
$q_{\text{rxn}} < 0$ confirms this is an **exothermic** process (dissolving NaOH releases heat).

Bomb Calorimetry

- Sealed, rigid vessel — measures at **constant volume**.
- Specifically used for combustion reactions.
- The bomb has a known heat capacity C_{bomb}:

$$q_{\text{rxn}} = -C_{\text{bomb}}\,\Delta T$$

- Measures ΔE (internal energy), not ΔH.
- $\Delta H \approx \Delta E$ for most combustion reactions.



Coffee-cup: ΔH; bomb: ΔE.

$\Delta H \approx \Delta E$ for most combustion reactions.

Check Your Understanding

When 1.00 g of benzoic acid is burned in a bomb calorimeter, the temperature rises by 2.57 °C. The calorimeter has a heat capacity of $3.34\,\frac{\mathrm{kJ}}{^\circ\mathrm{C}}$.

What is q_{rxn} (the heat of the reaction)?

Come to lecture to see the answer.

Part III

Enthalpy

Outline of Part III: Enthalpy

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Enthalpy (H)

Definition: Enthalpy

Enthalpy (H) is a thermodynamic state function defined as:

$$H = E + PV$$

where E is internal energy, P is pressure, and V is volume.

At **constant pressure**:

$$\begin{aligned} \Delta H &= \Delta E + P\Delta V \\ &= \Delta E - w \quad (\because w = -P\Delta V) \\ &= (q + w) - w \\ \therefore \quad \Delta H &= q_P \end{aligned}$$

The enthalpy change equals the **heat exchanged at constant pressure**.

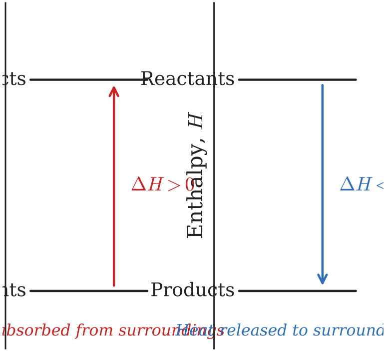
Endothermic and Exothermic Reactions

Endothermic ($\Delta H > 0$)

System **absorbs** heat.

Products have *more* enthalpy.

E.g., melting ice, photosynthesis



Exothermic ($\Delta H < 0$)

System **releases** heat.

Products have *less* enthalpy.

E.g., combustion, dissolving NaOH

$$\Delta H > 0 \Rightarrow \text{endothermic} \qquad \Delta H < 0 \Rightarrow \text{exothermic}$$

Check Your Understanding

Classify each process as **endothermic** or **exothermic**:

(a) Burning wood in a fireplace

(b) An ice pack becoming cold when activated

(c) Condensation of water vapor on a cold glass

(d) Baking a cake

Come to lecture to see the answer.

Outline of Part III: Enthalpy

Thermochemical Equations

Definition

A **thermochemical equation** is a balanced chemical equation that also states the enthalpy change ΔH of the reaction.

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(\ell) \qquad \Delta H = -572\,\mathrm{kJ}$$

📝 Notes on Thermochemical Equations:

- ΔH refers to the equation **as written** (with the given stoichiometric coefficients).
- State symbols (g, l, s, aq) are **required** — enthalpy depends on phase.
- Reversing the reaction **changes the sign** of ΔH.
- Multiplying the equation by a factor **multiplies** ΔH by the same factor.

Thermochemical Equations: Reverse and Scale

Original

$$2\,H_2O(\ell) \longrightarrow 2\,H_2(g) + O_2(g) \qquad \Delta H = 572\,\mathrm{kJ}$$

Reversed (sign of ΔH changes)

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(\ell) \qquad \Delta H = -572\,\mathrm{kJ}$$

Scaled by $\frac{1}{2}$ (ΔH also scaled by $\frac{1}{2}$)

$$H_2(g) + \tfrac{1}{2}\,O_2(g) \longrightarrow H_2O(\ell) \qquad \Delta H = -286\,\mathrm{kJ}$$

Check Your Understanding

Given:

$$C(s) + O_2(g) \longrightarrow CO_2(g) \qquad \Delta H = -393.5\,\mathrm{kJ}$$

What is ΔH for the following reaction?

$$3\,CO_2(g) \longrightarrow 3\,C(s) + 3\,O_2(g)$$

Come to lecture to see the answer.

Outline of Part III: Enthalpy

Standard Enthalpy of Combustion

Definition

The **standard enthalpy of combustion** (ΔH_c°) is the enthalpy change when 1 mol of a substance burns completely in oxygen under standard conditions (25 °C, 1 atm).

Substance	ΔH_c° ($\frac{\text{kJ}}{\text{mol}}$)
Hydrogen, $H_2(g)$	−286
Carbon (graphite), C(s)	−394
Methane, $CH_4(g)$	−890
Propane, $C_3H_8(g)$	−2220
Ethanol, $C_2H_5OH(l)$	−1367

Combustion enthalpies are always **negative** (exothermic).

Part IV

Hess's Law & Standard Enthalpies

Outline of Part IV: Hess's Law & Standard Enthalpies

Hess's Law

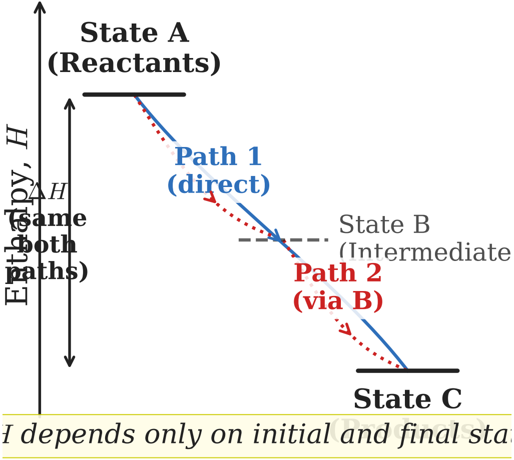
Standard Enthalpy of Formation

Outline of Part IV: Hess's Law & Standard Enthalpies

Hess's Law

Standard Enthalpy of Formation

State Functions and Path Independence



Enthalpy is a State Function

The value of a **state function** depends only on the **initial and final states** of a system, *not* on the path taken between them.

Energy, temperature, and enthalpy are state functions.
Heat (q) and work (w) are *not*.

Analogy: The elevation change when hiking from a valley to a summit is the same regardless of which trail you take. Elevation is a state function; the distance walked is not.

Hess's Law

Hess's Law

If a reaction can be expressed as the sum of two or more reactions, then ΔH for the overall reaction equals the **sum** of the ΔH values of those reactions.

Procedure

1. Identify reactions whose sum gives the target reaction.
2. **Reverse** any reaction as needed (change sign of ΔH).
3. **Scale** any reaction as needed (multiply ΔH by same factor).
4. Add the reactions and their ΔH values; cancel species that appear on both sides.

Example: Hess's Law

Finding ΔH for $4\,NO_2(g) + O_2(g) \rightarrow 2\,N_2O_5(g)$

Given:

(a) $N_2O_5(g) \rightarrow 2\,NO_2(g) + \frac{1}{2}\,O_2(g) \qquad \Delta H_a = 223.7\,\text{kJ}$

(b) $NO(g) + \frac{1}{2}\,O_2(g) \rightarrow NO_2(g) \qquad \Delta H_b = -57.1\,\text{kJ}$

Strategy: reverse (a) twice; reverse (b) four times.

$-2\times$(a): $4\,NO_2 + O_2 \rightarrow 2\,N_2O_5 \qquad \Delta H = 2 \times (-223.7) = -447.4\,\text{kJ}$

$$\begin{aligned}\Delta H_{\text{rxn}} &= 2(-\Delta H_a) + 4(-\Delta H_b)\\ &= 2(-223.7) + 4(-(-57.1))\end{aligned}$$

$$\boxed{\Delta H_{\text{rxn}} = -447.4 + 228.4 = -219.0\,\text{kJ}}$$

Check Your Understanding

Use Hess's Law to find ΔH for:

$$C(s) + \tfrac{1}{2}\,O_2(g) \rightarrow CO(g)$$

Given:

(a) $C(s) + O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -393.5\,\text{kJ}$

(b) $CO(g) + \tfrac{1}{2}\,O_2(g) \rightarrow CO_2(g)$ $\quad \Delta H = -283.0\,\text{kJ}$

Come to lecture to see the answer.

Outline of Part IV: Hess's Law & Standard Enthalpies

Standard Enthalpy of Formation (ΔH_f°)

Definition

The **standard enthalpy of formation** (ΔH_f°) is the enthalpy change when 1 mol of a compound is formed from its **constituent elements in their standard states** at 25 °C and 1 atm.

$$\mathrm{C(s,\ graphite)} + 2\,\mathrm{H_2(g)} \longrightarrow \mathrm{CH_4(g)} \qquad \Delta H_f^\circ = -74.6\,\frac{\mathrm{kJ}}{\mathrm{mol}}$$

By definition, the standard enthalpy of formation of any **element in its standard state** is exactly **zero**:

$$\Delta H_f^\circ[\mathrm{O_2(g)}] = 0, \quad \Delta H_f^\circ[\mathrm{C(s,\ graphite)}] = 0, \quad \Delta H_f^\circ[\mathrm{Fe(s)}] = 0$$

Standard Enthalpies of Formation: Selected Values

Compound	ΔH_f° $\left(\frac{\text{kJ}}{\text{mol}}\right)$
$H_2O(l)$	−286.8
$H_2O(g)$	−241.8
$CO_2(g)$	−393.5
$CH_4(g)$ (methane)	−74.6
$NH_3(g)$	−46.1
$NO_2(g)$	+33.2
$C_6H_6(l)$ (benzene)	+49.1

Most compounds have *negative* ΔH_f° — they are thermodynamically more stable than their elements. A *positive* ΔH_f° (e.g. NO_2, benzene) means the compound is less stable than its elements.

Calculating $\Delta H^\circ_{\text{rxn}}$ from Formation Enthalpies

Hess's Law Applied to Formation Enthalpies

$$\Delta H^\circ_{\text{rxn}} = \sum n \Delta H^\circ_f(\text{products}) - \sum m \Delta H^\circ_f(\text{reactants})$$

where n and m are stoichiometric coefficients.

Strategy: "products minus reactants," each multiplied by its stoichiometric coefficient.

Mnemonic

$$\Delta H^\circ_{\text{rxn}} = \sum(\text{products}) - \sum(\text{reactants})$$

"What you end up with minus what you started with."

Example: $\Delta H^\circ_{\text{rxn}}$ from ΔH°_f

Combustion of Methane

Calculate $\Delta H^\circ_{\text{rxn}}$ for:

$$\mathrm{CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(\ell)}$$

Use: $\Delta H^\circ_f[\mathrm{CH_4(g)}] = -74.6\,\frac{\text{kJ}}{\text{mol}}$, $\Delta H^\circ_f[\mathrm{CO_2(g)}] = -393.5\,\frac{\text{kJ}}{\text{mol}}$, $\Delta H^\circ_f[\mathrm{H_2O(\ell)}] = -286.8\,\frac{\text{kJ}}{\text{mol}}$, $\Delta H^\circ_f[\mathrm{O_2(g)}] = 0$

$$\begin{aligned}\Delta H^\circ_{\text{rxn}} &= \big[1(-393.5) + 2(-286.8)\big] - \big[1(-74.6) + 2(0)\big] \\ &= [-393.5 - 573.6] - [-74.6]\end{aligned}$$

$$\boxed{\Delta H^\circ_{\text{rxn}} = -892.5\,\text{kJ}}$$

Check Your Understanding

Calculate $\Delta H^\circ_{\text{rxn}}$ for the formation of water vapor from its elements:

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(g)$$

Use $\Delta H^\circ_f[H_2O(g)] = -241.8\,\frac{\text{kJ}}{\text{mol}}$.

Come to lecture to see the answer.

Check Your Understanding

Rank these processes from **most endothermic** to **most exothermic** (no calculation needed):

(a) Burning hydrogen: $\Delta H = -286\,\frac{\text{kJ}}{\text{mol}}$

(b) Dissolving ammonium nitrate: $\Delta H = 25\,\frac{\text{kJ}}{\text{mol}}$

(c) Combustion of propane: $\Delta H = -2220\,\frac{\text{kJ}}{\text{mol}}$

(d) Formation of NO_2: $\Delta H = 33\,\frac{\text{kJ}}{\text{mol}}$

Come to lecture to see the answer.

Summary: Chapter 5

- **Energy** exists in many forms; transfers appear as heat (q) or work (w).
- **First Law:** $\Delta E = q + w$; energy is conserved; $\Delta E_{\text{universe}} = 0$.
- **PV work:** $w = -P\Delta V$; expansion does work on surroundings ($w < 0$).
- **Specific heat:** $q = mc\Delta T$; water has an unusually high c ($4.184\,\frac{\text{J}}{\text{g}\,^\circ\text{C}}$).
- **Calorimetry:** $q_{\text{rxn}} = -q_{\text{calorimeter}}$; coffee-cup measures ΔH; bomb measures ΔE.
- **Enthalpy:** $H = E + PV$; at constant pressure, $\Delta H = q_P$.
- **Endothermic** ($\Delta H > 0$) vs. **Exothermic** ($\Delta H < 0$).
- **Hess's Law:** ΔH is a state function; ΔH_{rxn} is path-independent.
- **Standard enthalpy of formation** (ΔH_f°): enthalpy to form 1 mol of compound from elements in standard states; elements have $\Delta H_f^\circ = 0$.
- $\Delta H^\circ_{\text{rxn}} = \sum n\,\Delta H_f^\circ(\text{products}) - \sum m\,\Delta H_f^\circ(\text{reactants})$.